Quiz # 4 (Data Structures) CLO2 **BS DS/CY Fall 2025**

Name:_____ **Roll number:_____**
Time Allowed: 20 min **Total Marks: 20**

Q1: Write down the output of the following codes:

`int fun1(int a[], int n) {`
`  if (n == 1)`
`  return a[0];`
`  else {`
`  int x = fun1(a, n - 1);`
`  if (x < a[n - 1])`
`  return x;`
`  else`
`  return a[n - 1];`
`  }`
`}`
`int main()`
`{`
`  int arr[] = {12, 10, 30, 50, 100};`
`  cout << fun1(arr, 5) << endl;`
`  return 0;`
`}`
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`int fun(int n) {`
`  if (n > 100)`
`  return n - 5;`
`  return fun(fun(n + 11));`
`}`
`int main() {`
`  cout << fun(65) << endl;`
`  return 0;`
`}`
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`int test(int i, int j) {`
`  if (i == 0)`
`  return j;`
`  else`
`  return test(i - 1, i * j);`
`}`
 	
`int main()`
`{`
`  cout << test(4, 2) << endl;`
`  return 0;`
`}`
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`void fun4(int n) {`
`  if (n > 0) {`
 	
`  fun4(n - 1);`
`  fun4(n - 1);`
`  cout << n << " ";`
`  }`
`}`
`int main()`
`{`
`  fun4(3);`
`  return 0;`
`}`
	1 1 2 1 1 2 3